



Z(3rd Sm.)-Computer Sc.-G/(GE/CC-3)/CBCS

2023

COMPUTER SCIENCE — GENERAL

Paper : GE/CC-3

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
 - (a) Write down any two characteristics of RISC architecture.
 - (b) State the role of accumulator.
 - (c) An 8 bit register contains the binary value 10111101. What is the register value after an arithmetic shift right?
 - (d) What is BUS Arbitration?
 - (e) Write down the functionality of addressing modes.
 - (f) What is the function of cache memory?
2.
 - (a) What do you mean by Instruction set completeness?
 - (b) Evaluate the following arithmetic expression : $X = (P - Q + R) / S * T$ using zero address instruction format. 5+5
3.
 - (a) Explain the concept of Direct Memory access with suitable illustration.
 - (b) State the differences between direct and indirect addressing mode. 5+5
4.
 - (a) A computer employs RAM chips of 256×8 and ROM chips of 1024×8 . The computer system needs 2K bytes of RAM, 4K bytes of ROM. How many number of RAM and ROM chips are needed?
 - (b) Consider a direct mapped cache of size 32 KB with block size 512 Bytes. The size of main memory is 16 MB. Find the number of bits in tag and line number.
 - (c) What is virtual memory? 3+5+2
5.
 - (a) Explain different types of cache write policies.
 - (b) Write down the significances of Priority Interrupt.
 - (c) Describe the functions of system bus. 5+2+3

6. (a) Write short note on memory hierarchy with appropriate illustrations. 5+5
(b) Define prefix, postfix and infix instruction with example. 5+5
7. (a) Highlight the main differences between Hardwired control and Microprogrammed control unit. 5+5
(b) Differentiate between direct and indirect addressing. 5+5
8. (a) Illustrate the functions of associative memory. 5+5
(b) Write down the characteristics of USB and PCI. 5+5
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